

Pentium 4 NetBurst Microarchitecture — MPCA Module 6

Definition: The **Intel NetBurst Microarchitecture** is a 7th-generation x86 processor design engineered to achieve ultra-high clock speeds (up to 3.8 GHz) and maximum throughput using a deep **20-stage (later 31-stage Prescott) Hyper-Pipelined Technology**.

1. Detailed Technical Explanation

Functional Components of NetBurst Microarchitecture

NETBURST MICROARCHITECTURE

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| [ Advanced Dynamic Execution Engine ] |
|   - Out-of-Order Execution Logic      |
|   - 126 In-Flight Instructions Window |
|                                       |
| [ Execution Trace Cache ]             |
|   - Stores 12,000 decoded micro-ops (uops) |
|   - Bypasses traditional instruction decoder on loops & branch hits |
|                                       |
| [ Rapid Execution Engine (ALUs) ]     |
|   - Integer ALUs clocked at TWICE (2x) the core processor frequency |
|   - Executes simple arithmetic (ADD, SUB) in 0.5 clock cycles! |
|                                       |
| [ High-Performance System Bus ]      |
|   - Quad-Pumped Front Side Bus (FSB) at 400 / 533 / 800 MHz (6.4 GB/s) |
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2. Deep Dive into NetBurst Innovations

1. Execution Trace Cache (L1 I-Cache Replacement)

- Traditional architectures store raw x86 instructions in L1 instruction cache, requiring repeated decode cycles.
- Trace Cache** stores **pre-decoded micro-operations (uops)** in their predicted execution path order.
- Decodes instructions *before* caching, removing the x86 instruction decoder bottleneck during execution loops.

2. Hyper-Pipelined Technology (20 to 31 Stages)

- Decomposes instruction processing into 20 distinct pipelined clock stages.
- Enables very high operating frequencies (over 3.0 GHz) by reducing the amount of logic executed per stage.

3. Rapid Execution Engine (2x Clock ALUs)

- Integer Arithmetic Logic Units (ALUs) run at **double the core processor frequency**.
- Basic integer operations like **ADD**, **SUB**, and bitwise logic complete in **half a clock cycle**.

4. Quad-Pumped System Bus

- Transfers data 4 times per clock cycle over a 64-bit bus, delivering up to **6.4 GB/s memory bandwidth** at 800 MHz FSB.
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3. Core Concepts & Memory Keywords

- **NetBurst:** 7th-gen x86 design focused on high clock frequencies.
 - **Trace Cache:** Caches 12K decoded micro-ops instead of raw x86 bytes.
 - **Rapid Execution Engine:** 2x core clock rate ALUs executing in 0.5 cycles.
 - **Hyper-Pipeline:** 20-stage pipeline depth.
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4. Quick Recall Flow

NetBurst -> 20-Stage Hyper-Pipeline -> Execution Trace Cache (12K uops) -> Rapid Engine (2x ALUs) -> Quad-Pumped FSB (800MHz)